

A randomized trial of the effects of two types of short-term hypocaloric diets on weight loss in women with polycystic ovary syndrome.

OBJECTIVE: We performed this study as a pilot experiment to investigate the short term effects of two diets of varying composition on weight loss as the primary outcome in obese women with polycystic ovary syndrome (PCOS) seeking fertility. **DESIGN:** Randomized clinical trial. **SETTING:** Academic medical center. **PATIENT(S):** Thirty-five obese women with PCOS. **INTERVENTION(S):** We examined the effects of a 1-month dietary intervention on the PCOS phenotype. Participants were randomized to one of two energy-restricted diets; high protein (HP: 30% protein, 40% carbohydrate, and 30% fat) or high carbohydrate (HC: 15% protein, 55% carbohydrate, and 30% fat). The fat content was held constant in both diets. **MAIN OUTCOME MEASURE(S):** Primary - change in body weight; Secondary - biometric, hormonal, lipid and lipoprotein, and markers of glucose homeostasis and energy metabolism. **RESULT(S):** Twenty-six women completed the study. Both the HP (-3.7 +/- 1.9 kg) and HC (-4.4 +/- 1.5 kg) diets resulted in significant weight loss, but there was no significant difference in mean weight loss between the two groups. There were also no differences between diets on a variety of measures including circulating androgens, measures of glucose metabolism, and leptin. However, the effects of a hypocaloric diet per se on improving metabolic and reproductive abnormalities in a group of PCOS women were marked by a decline in circulating androgens ($P=.03$), fasting and area under the curve (AUC) insulins ($P<.05$) on a 3-hour oral glucose tolerance test (OGTT), and fasting and AUC leptin levels ($P<.0001$). There was a high prevalence of menstrual bleeding during the trial (14 out of 26 patients). **CONCLUSION(S):** Those who completed the short-term hypocaloric diet had a significant weight loss and a significant improvement in their reproductive and metabolic abnormalities. There was no increased benefit to a high-protein diet. Future diet studies evaluating the ideal composition of a hypocaloric diet in women with PCOS will require a large study population, and will most likely require a multicenter trial.